

Bwabh Alabtkar (Innovation Hub) — System Documentation Package

*Note: This documentation package accompanies the **Bwabh Alabtkar** enterprise innovation and workforce skill management platform. It is structured for both executive decision-makers and technical reviewers evaluating the system.*

Table of Contents

- 1. Executive Summary
- 2. Features & Capabilities
- 3. User Guide & Workflows
- 4. Seeded Demo Accounts
- 5. Architecture Overview
- 6. Technology Stack & Design Rationale
- 7. Deployment & Demonstration Environment Notes
- 8. Testing & Evaluation Instructions
- 9. Current System Limitations
- 10. Future Production Roadmap
- 11. Appendix: API Endpoints & Reference

1. Executive Summary

What the Application Does

Bwabh Alabtkar (■■■■■ ■■■■■■■■ — *Innovation Hub*) is a comprehensive enterprise SaaS platform designed to bridge workforce skill gaps and manage organizational innovation. The platform provides an end-to-end ecosystem where organizations can collect workforce ideas, evaluate skill competencies, recommend personalized development pathways, and track training impact on organizational performance.

The Problem It Solves

Traditional organizations struggle with two core challenges:

- 1. **Workforce Skill Fragmentation:** Training programs are often generic, lacking alignment with actual employee skill deficiencies and strategic business goals.
- 2. **Unorganized Innovation Intake:** Valuable ideas from front-line employees are buried in emails or informal channels, missing structured evaluation and execution.

Bwabh Alabtkar consolidates skill assessment, OTP-secured authentication, multi-role administration, and structured innovation intake into a single web application.

Target Users

- **Super Administrators:** Executive IT/HR heads managing administrative roles and system configuration.
- **Standard Administrators:** Operations managers overseeing customer accounts, user onboarding, and portal management.
- **Corporate Customers & Employees:** Enterprise staff submitting innovation proposals, verifying credentials, and accessing skill development resources.

Business Value

- **Higher ROI on Training:** Data-driven skill gap analysis ensures training budgets target verified competency deficiencies.
 - **Accelerated Innovation Lifecycle:** Reduces idea evaluation time from weeks to days via centralized intake workflows.
 - **Enterprise-Grade Security:** Secured with stateless JWT authentication, Spring Security, Redis-backed OTP state management, and Jakarta Bean Input Validation (@Valid).
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2. Features & Capabilities

2.1 Customer Portal Features

- **Responsive Landing Page:** Multi-device optimized presentation highlighting workforce skill bridging and innovation management.
- **Industry Solutions Showcase:** Specialized solution modules tailored for **Education, Enterprise, Government,** and **Healthcare** sectors.
- **Product Suite:** Interactive pages for **Whiteboard Collaboration, Idea Management,** and **AI Skill Analysis.**
- **Secure Authentication Suite:**
- **Account Registration:** Full name, email, and password registration with strict frontend/backend validation.
- **Single-Click OTP Verification:** 6-digit email verification delivered via Resend HTML API, featuring **one-click paste** that automatically splits 6-digit clipboard codes across input boxes.
- **Stateless JWT Sign-in:** Secure authentication returning bearer JWT tokens.
- **Password Recovery Flow:** Forgot password workflow with email verification code and multi-step verification before password update.

2.2 Admin Management Dashboard

- **Material Dashboard Aesthetic:** Sleek dark/light theme built on Material Dashboard with custom Toast notifications and confirmation dialogs.
 - **Administrator Management:**
 - Create new Standard Admins or Super Admins.
 - Edit administrator details with email immutability enforcement.
 - Soft-deactivate and reactivate administrator accounts with Material dialog confirmation.
 - **Customer Management:**
 - Monitor customer status (Active / Deactivated).
 - Update customer profiles and manage account lifecycle.
 - **Navbar Profile Dropdown & Session Management:**
 - Topbar user dropdown displaying active user name and email.
 - One-click secure Logout clearing session tokens and returning to sign-in.
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3. User Guide & Workflows

3.1 Customer Workflow (Sign-Up & Verification)

1. **Navigate to Sign-Up:** Click **Create Account** on the sign-in page (<docs/auth/signup.html>).
2. **Enter Credentials:** Fill in Full Name, Email, and Password (minimum 6 characters).
3. **Email OTP Delivery:** The system generates a 6-digit verification code stored in Redis and sends a styled HTML email to your inbox.

4. **One-Click OTP Entry:** Copy the 6-digit code from your email, click the first OTP box, and paste (**Ctrl+V** or right-click paste). All 6 boxes populate automatically.
5. **Complete Account Activation:** Click **Verify Code** to activate your account and proceed to Sign-In.

3.2 Password Reset Workflow

- Click **Forgot Password?** on the sign-in page (<docs/auth/forgot-password.html>).
- Enter your registered email address and submit.
- Check your email for the **Bwabh Alabtkar Password Reset Code**.
- Paste the 6-digit code on the verification screen (<docs/auth/verify-reset.html>).
- Set your new password on the final screen (<docs/auth/reset-password.html>).

3.3 Admin Workflow (Dashboard Management)

- Navigate to <docs/admin/pages/dashboard.html> and sign in using an Admin account.
- View System Analytics:** Access customer overview and system status metrics.
- Manage Administrators** (<docs/admin/pages/admins.html>):
 - Click **Add New Administrator** to open the modal.
 - Enter Name, Email, Password, and select Role (**ROLE_ADMIN** or **ROLE_SUPER_ADMIN**).
 - Click **Save** to submit. The system validates inputs via **@Valid** and displays toast notifications.
- Deactivate/Reactivate Accounts:**
 - Click **Deactivate** on an active account row.
 - Confirm the action in the custom Material modal.
 - To restore access, click **Reactivate**.
- Logout:** Click the user profile icon ([account_circle](#)) in the top-right navbar and click **Logout**.

4. Seeded Demo Accounts

The database contains pre-configured demonstration accounts for system evaluation:

Account Type	Role Granted	Email Address	Password	Key Permissions & Capability Scope	Recommended Testing Scenarios
Super Admin	ROLE_SUPER_ADMIN ROLE_ADMIN	superadmin@bwabhalabtkar.com	admin123	• Full System Control • Create & Edit Administrators • Deactivate / Reactivate Admins • Manage Customer Accounts • View Dashboard Metrics	Test admin creation, role assignments, admin deactivation/reactivation, and customer management.

Standard Admin	ROLE_ADMIN	admin@bwabhalabtkar.com	admin123	• Operational Management • Manage Customer Accounts • View Dashboard Metrics • Restricted from editing Super Admins	Test standard admin navigation, customer deactivation, and restricted super admin boundaries.
Customer User	ROLE_CUSTOMER	customer@bwabhalabtkar.com	password123	• Portal Access • Explore Solutions & Products • Profile & Password Recovery	Test sign-in, solution pages exploration, and password reset workflow.

5. Architecture Overview

Key Components

- **Frontend Layer:** Static web package stored in docs/ (GitHub Pages compatible), built with Vanilla HTML5, CSS3 (WebP media optimization), and ES modules.
- **Backend Layer:** Spring Boot 4.1.0 application with Java 26, Spring Data JPA, and Spring Validation (@Valid Bean Validation).
- **Database & Cache:** Relational storage for permanent user entities, paired with Redis in-memory cache for ephemeral OTP registration states (PendingSignup, PendingPasswordReset) with 10-minute automatic expiration.
- **Security & Authentication:** Stateless JWT bearer tokens (io.jsonwebtoken), BCrypt password hashing, and Spring Security method-level protection.
- **Email Delivery:** Integrated with the Resend API for HTML transactional email delivery.

6. Technology Stack & Design Rationale

Layer	Technology	Version	Rationale & Selection Criteria
Backend Framework	Spring Boot	4.1.0 (Java 26)	Provides robust enterprise Java infrastructure, dependency injection, and native security filters.
Security & Auth	Spring Security + JWT	0.12.5	Enables stateless token-based authentication with zero server session overhead.
Validation	Jakarta Bean Validation	3.1.0	Enforces strict DTO constraints (@NotBlank, @Email, @Pattern, @Size) at controller endpoints.
Cache Store	Redis	7.x	In-memory key-value cache with automatic TTL expiration for secure 6-digit OTP codes.
Transactional Email	Resend Java SDK	3.1.0	High-deliverability API for rendering HTML verification emails.
Frontend Core	HTML5 / CSS3 / Vanilla JS	ES6+	Zero-dependency browser execution ensuring ultra-fast load times and lightweight footprint.

Media Format	WebP	Lossless 85%	Replaced PNG/JPG files, reducing image payload size by ~80% (~29 MB saved).
Admin UI Theme	Material Dashboard	3.2.0	High-end dashboard component framework with responsive sidebar and table controls.

7. Deployment & Demonstration Environment Notes

Demonstration Environment Notice:

- The live backend service for this evaluation environment is hosted locally on a development machine and exposed securely using a **reverse HTTP/HTTPS tunnel**.
- Because API requests travel over a local broadband connection through tunnel relays, response times may experience minor latency compared to cloud deployment.
- In a **production deployment** (e.g., AWS EC2/ECS, GCP, or a dedicated Cloud VPS with managed Redis and PostgreSQL), response times are sub-100ms.

8. Testing & Evaluation Instructions

To evaluate the application thoroughly, test the following core scenarios:

Scenario A: New User Registration & Single-Click OTP Paste

1. Open `docs/auth/signup.html`.
2. Enter your name, email, and password, then click **Create Account**.
3. Check your email for the 6-digit code.
4. Copy the code, click the first OTP input box, and press `Ctrl+V`. Verify that all 6 boxes populate instantly.
5. Click **Verify Code** and confirm account creation.

Scenario B: DTO Input Validation (Backend Enforcement)

1. Open `docs/auth/login.html` and attempt to submit an invalid email format (e.g., `user@test`).
2. Observe backend validation response returning structured JSON field errors.
3. Test short password entry on signup to verify `@Size(min = 6)` enforcement.

Scenario C: Admin Management & Role Actions

1. Open `docs/admin/pages/admins.html` and log in as `superadmin@bwabhalabtkar.com` (`admin123`).
2. Click **Add New Administrator** and create a new Standard Admin.
3. Deactivate an administrator account and verify the Material modal confirmation dialog.
4. Log out via the top-right account dropdown.

9. Current System Limitations

PoC Notice:

The current system is deployed as a **Proof of Concept (PoC)**. The following limitations are intentional for demonstration scope:

1. **Reverse Tunnel Backend:** Local host backend connectivity subject to local network variations.

2. **Email Sandbox Mode:** Email delivery via Resend API operates under free tier rate limits (100 emails/day).
3. **Single Region Database:** Operating single-instance relational storage without multi-region read replicas.

10. Future Production Roadmap

For full commercial production readiness, the following enhancements are planned:

1. **Cloud Native Infrastructure:** Containerize backend services with Docker & Kubernetes (EKS/GKE) behind an AWS Application Load Balancer.
2. **CI/CD Automation:** Automated testing, linting, and deployment pipelines via GitHub Actions.
3. **Observability & Monitoring:** Spring Boot Actuator integration with Prometheus & Grafana dashboards.
4. **Enhanced Content Security:** Strict Content Security Policy (CSP), CORS domain restriction, and Rate Limiting filters ([Bucket4j](#) / Redis rate limiter).
5. **SSO Integration:** OAuth2 / SAML 2.0 integration for enterprise Single Sign-On (Google Workspace, Microsoft Entra ID).

11. Appendix

11.1 Live Demonstration URLs

- **Customer Portal:** [docs/customer/pages/home.html](#) (GitHub Pages)
- **Admin Dashboard:** [docs/admin/pages/dashboard.html](#)
- **Sign-In Page:** [docs/auth/login.html](#)

11.2 API REST Endpoint Summary

HTTP Method	Endpoint	Description	Auth Required	Validation DTO
POST	/api/auth/signup	Initiate user registration & send OTP	No	SignupRequest.java
POST	/api/auth/login	Authenticate user & return JWT	No	LoginRequest.java
POST	/api/auth/verify-otp	Verify 6-digit registration code	No	OtpVerificationRequest.java
POST	/api/auth/forgot-password	Send password reset OTP	No	ForgotPasswordRequest.java
POST	/api/auth/reset-password	Set new password with OTP	No	ResetPasswordRequest.java
GET	/api/admin/management	List all administrators	Yes (ROLE_ADMIN)	N/A
POST	/api/admin/management	Create new administrator	Yes (ROLE_SUPER_ADMIN)	AdminCreateRequest.java
DELETE	/api/admin/management/{id}	Soft-deactivate administrator	Yes (ROLE_SUPER_ADMIN)	N/A
PATCH	/api/admin/management/{id}/reactivate	Reactivate administrator	Yes (ROLE_SUPER_ADMIN)	N/A